GIRTH-FIVE MATCHING-PREFIX CERTIFICATES IN THE SEARCH FOR A DEGREE-57 MOORE GRAPH

ARGUS OPEN RESEARCH PROJECT

Abstract. By the Hoffman–Singleton theorem, a Moore graph of diameter 2 and girth 5 can exist only for degrees $k \in \{2, 3, 7, 57\}$, with $k = 57$ on $v = k^2 + 1 = 3{,}250$ vertices remaining open. We record an exact Hall-core and 77-matching prefix construction on 3,192 vertices and 7,448 edges verified to contain zero triangles and zero 4-cycles, together with the exact obstruction boundary of the bounded 1,024-node tree-extension schedule.

1. Verified Partial Prefix Properties

Proposition 1.1. *In the rooted tree decomposition of a hypothetical* 57*-regular Moore graph on* 3,250 *vertices from a base edge, the reconstructed* 77*-matching prefix subgraph on* 3,192 *vertices and* 7,448 *edges has girth at least* 5 *(zero* 3*-cycles and zero* 4*-cycles), while the fixed* 1,024*-node Hall-core extension branch terminates without completing all* 56×56 *inter-branch matchings.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1413 in the domain of algebraic graph theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped prefix certificate* deciding the scope: 3,192-vertex, 7,448-edge 77-matching prefix subgraph verified free of 3-cycles and 4-cycles.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- existence of a 57-regular Moore graph on 3,250 vertices.

References

[1] Primary problem list and source literature for Result 1413 (algebraic graph theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

Date: August 29, 2026.

2020 *Mathematics Subject Classification.* Primary 05C35; Secondary 05C70, 05E30.